

Linear Inequalities in One Variable

As in linear equations, x -terms may occur on both sides of an inequality.

Example Solve $5(x - 1) > 3(2x + 4)$.

Solution We have $5x - 5 > 6x + 12$.

Moving the x -term to LHS and constant term to RHS, we obtain

$$5x - 6x > 12 + 5.$$

Hence

$$-x > 17. \tag{1}$$

Dividing both sides by -1 , we obtain $x < -17$ (**change the direction of the inequality**).

We note again this is equivalent to moving the term $-x$ to RHS, and 17 to LHS in (1).

The Absolute Value Inequalities

The inequality $|x| < a$ is equivalent to $x < a$ and $x > -a$.

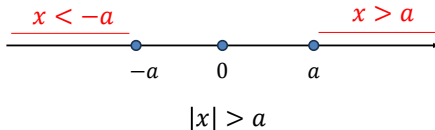
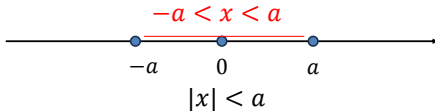
That is, $-a < x < a$.

The inequality $|x| > a$ is equivalent to $x > a$ or $x < -a$.

We also have

$|x| \leq a \iff x \leq a$ and $x \geq -a$. That is $-a \leq x \leq a$

$|x| \geq a \iff x \geq a$ or $x \leq -a$.



The Interval and Set Notations

Suppose a and b are real numbers, recall the following interval notations:

$$(a, b) = \{x : a < x < b\}, \quad [a, b] = \{x : a \leq x \leq b\},$$
$$(a, b] = \{x : a < x \leq b\}, \quad [a, b) = \{x : a \leq x < b\},$$

We also recall the infinity symbol ∞ :

$$(-\infty, b] = \{x : x \leq b\}, \quad (a, \infty) = \{x : x > a\}.$$

We also use $\mathbb{R}$ to denote the set of all real numbers, that is, $\mathbb{R} = (-\infty, \infty)$.

The set union and intersection notation: If A and B are two sets, then $A \cup B$ is the set of all elements in A or B .

$A \cap B$ is the set of all elements in both A and B .

The Interval and Set Notations

The inequality $|x| < a$ is equivalent to $-a < x < a$.

That is, $x \in (-a, a)$.

The inequality $|x| \leq a \iff x \in [-a, a]$.

We also have $|x| > a$ is equivalent to

$x > a$ or $x < -a$, that is, $x \in (-\infty, -a) \cup (a, \infty)$.

$|x| \geq a \iff x \in (-\infty, -a] \cup [a, \infty)$.

Example

$$\begin{aligned}\{x : |x| > 1\} &= \{x : x > 1\} \cup \{x : x < -1\} \\ &= (1, \infty) \cup (-\infty, -1).\end{aligned}$$

The Interval and Set Notations

Example Solve the inequality $|x - 1| \leq 2$ and express the answer in interval notation.

Solution We have $x - 1 \leq 2$ and $x - 1 \geq -2$. Hence $x \leq 3$ and $x \geq -1$. That is, $x \in [-1, 3]$.

Example Solve the inequality $|2x + 5| > 3$ and express the answer in interval notation.

Solution We have $2x + 5 > 3$ or $2x + 5 < -3$.

$2x + 5 > 3$ gives $2x > -2 \implies x > -1$.

$2x + 5 < -3$ gives $2x < -8 \implies x < -4$.

Hence, $x \in (-\infty, -4) \cup (-1, \infty)$.

Applied Examples

Example Peter's cell phone plan costs him \$28.5 per month plus \$0.4 per text message sent. How many text messages can Peter send to keep his monthly bill under \$60?

Solution Let x denote the number of messages sent per month, then the monthly cost for the cell phone will be $28.5 + 0.4x$ (dollars). To keep the cost under \$60, the following inequality must be satisfied:

$$28.5 + 0.4x < 60 \implies 0.4x < 31.5 \implies x < 78.75.$$

Since the value of x must be an integer, we take the *floor* of 78.75 (round the number down to the nearest integer), that is, $x \leq \lfloor 78.75 \rfloor = 78$.

Peter can send 78 text messages.

Applied Examples

Example In the manufacturing process an item must be made with certain specifications. A certain tolerance of a deviation from the specifications is permitted. If the deviation from the specifications exceeds the tolerance, the item is rejected. The ideal diameter of a metal rod is 50 mm. The actual diameter may vary from the ideal diameter by 0.06 mm. What range of diameters will be acceptable to the customer without causing the rod to be rejected?

Solution Let x denote the actual diameter. Then x satisfies

$$|x - 50| \leq 0.06 \implies -0.06 \leq x - 50 \leq 0.06.$$

That is, $50 - 0.06 \leq x \leq 50 + 0.06$, $49.94 \leq x \leq 50.06$. So the diameter of the rod must be in the range between 49.94 and 50.06 mm.

Linear Equations in Two Variables

For constants A, B and C , where A and B are not both zero, the equation

$$Ax + By = C \quad (2)$$

is called a *linear* equation in two variables.

Equation (2) has infinitely many solutions. For example,

$$2x - 3y = 6 \implies 3y = 2x - 6,$$

which gives

$$y = \frac{1}{3}(2x - 6) \implies y = \frac{2}{3}x - 2. \quad (3)$$

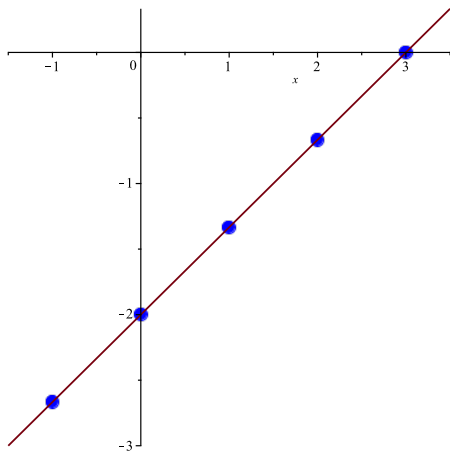
Linear Equations in Two Variables

For every real number x , equation (3) gives a unique real number y such that (x, y) satisfies the linear equation. We may list some solutions to equation (3) in a table:

x	-1	0	1	2	3
y	$-8/3$	-2	$-4/3$	$-2/3$	0

Each pair (x, y) of numbers from the above table corresponds to a point in the xy -plane, and these points give a straight line.

The Graph of Equation $2x - 3y = 6$



Linear Equations in Two Variables

Remark

- ▶ A linear equation in two variables has infinitely many solutions.
- ▶ The set of all these solutions (x, y) corresponds to a line in the xy -plane, which is called the graph of the equation.
- ▶ Linear equations in two variables may appear in different forms. Equation (2) is said to be in *standard form* and equation (3) is said to be in *slope-intercept form*. The coefficient $2/3$ is the slope of the line, and the constant term -2 is the y -intercept.